

Lecture Slide – 12

Newton's Divided Difference Interpolations

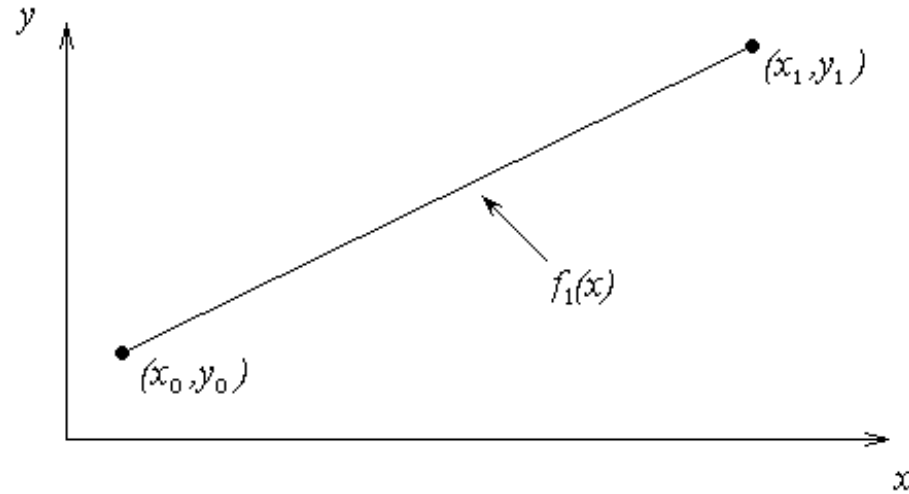
Newton's Divided Difference Method

Linear interpolation: Given (x_0, y_0) , (x_1, y_1) , pass a linear interpolant through the data

where $f_1(x) = b_0 + b_1(x - x_0)$

$$b_0 = f(x_0)$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$



Example

The upward velocity of a rocket is given as a function of time in Table 1. Find the velocity at $t=16$ seconds using the Newton Divided Difference method for linear interpolation.

Table 1. Velocity as a function of time

t (s)	$v(t)$ (m/s)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

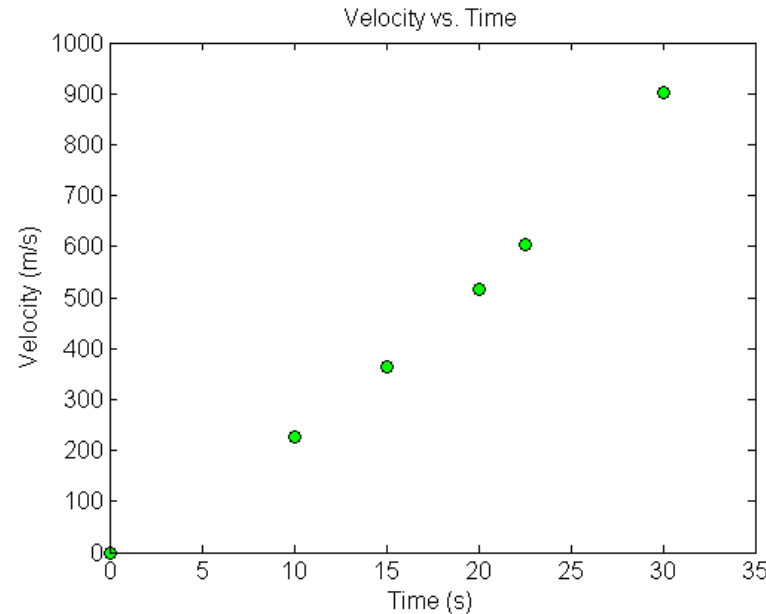


Figure. Velocity vs. time data for the rocket example



Linear Interpolation

$$v(t) = b_0 + b_1(t - t_0)$$

$$t_0 = 15, v(t_0) = 362.78$$

$$t_1 = 20, v(t_1) = 517.35$$

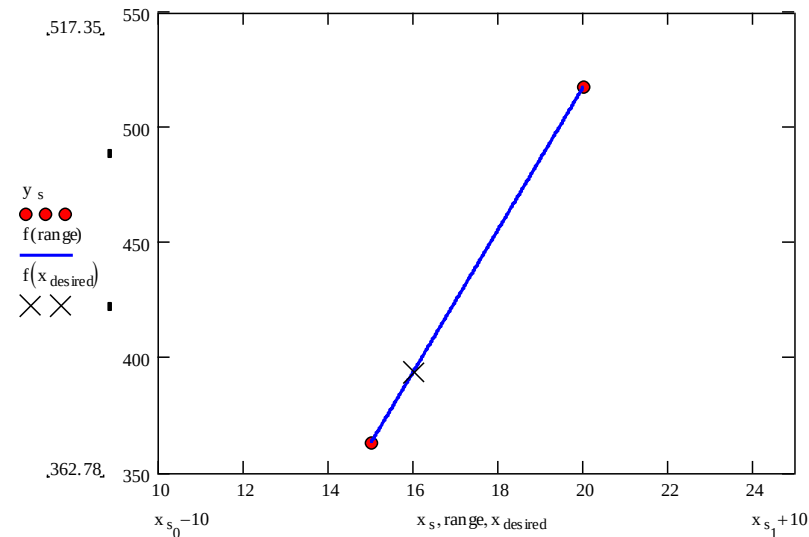
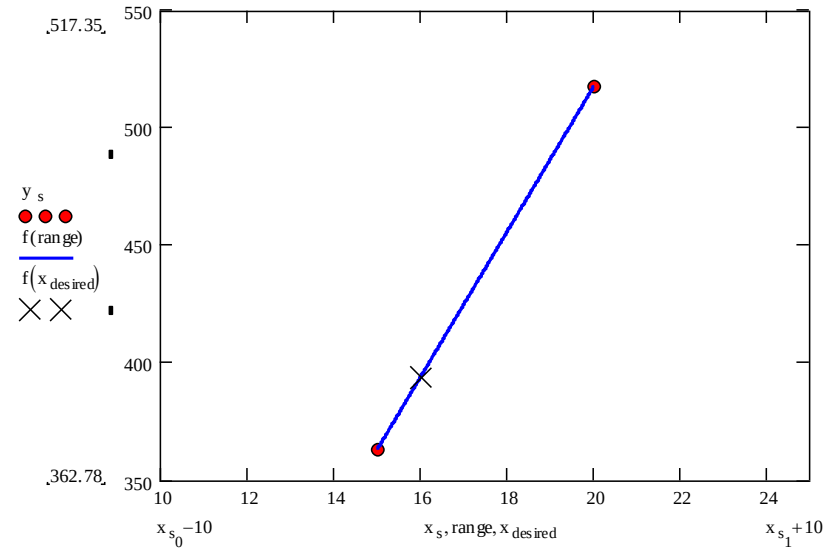
$$b_0 = v(t_0) = 362.78$$

$$b_1 = \frac{v(t_1) - v(t_0)}{t_1 - t_0} = 30.914$$

$$\begin{aligned} v(t) &= b_0 + b_1(t - t_0) \\ &= 362.78 + 30.914(t - 15), 15 \leq t \leq 20 \end{aligned}$$

At $t = 16$

$$\begin{aligned} v(16) &= 362.78 + 30.914(16 - 15) \\ &= 393.69 \text{ m/s} \end{aligned}$$



Quadratic Interpolation

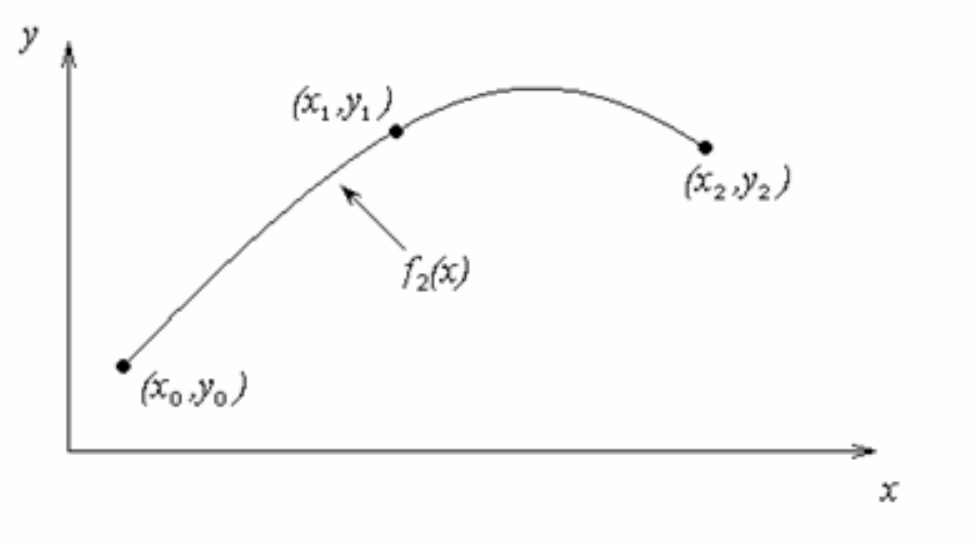
Given (x_0, y_0) , (x_1, y_1) , and (x_2, y_2) , fit a quadratic interpolant through the data.

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$b_0 = f(x_0)$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$



Example

The upward velocity of a rocket is given as a function of time in Table 2 Find the velocity at $t=16$ seconds using the Newton Divided Difference method for quadratic interpolation.

Table 2. Velocity as a function of time

t (s)	$v(t)$ (m/s)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

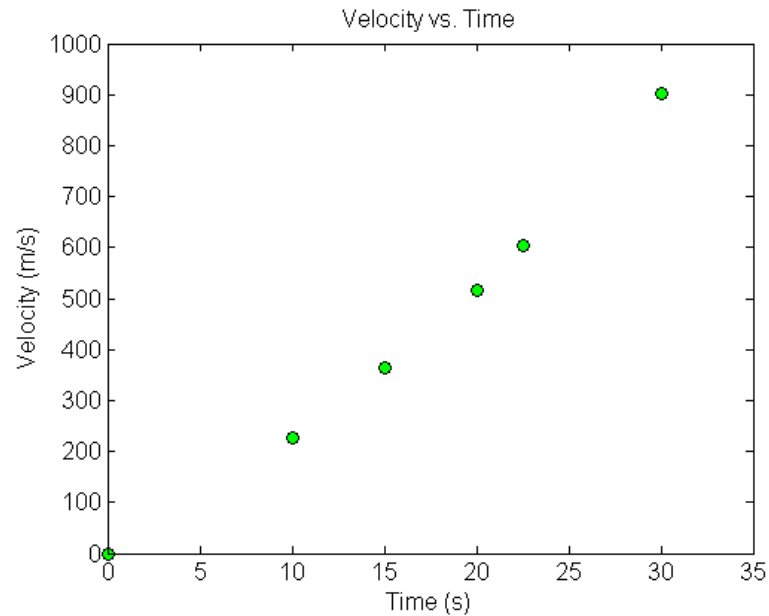
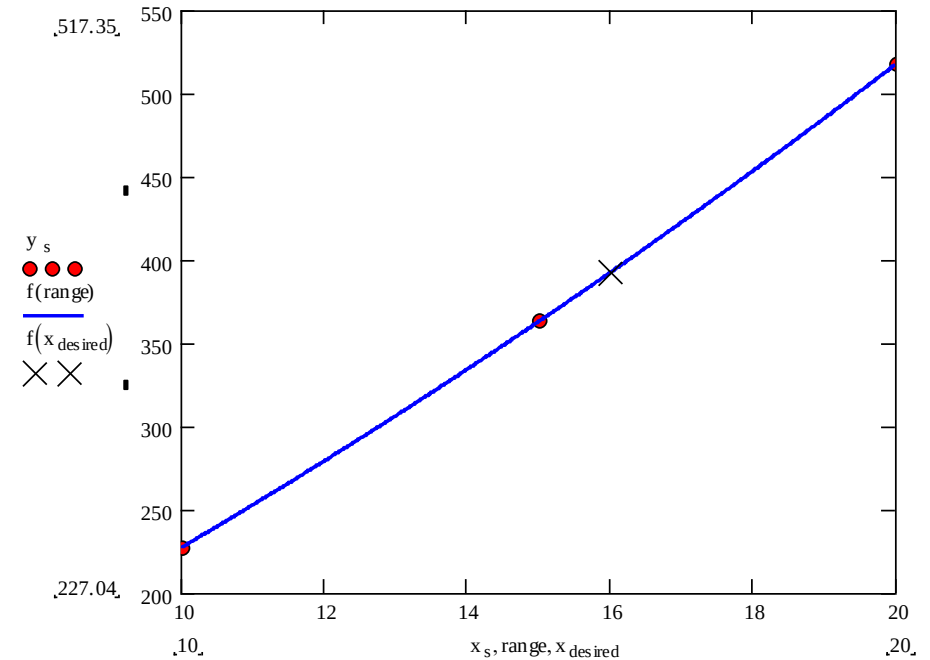


Figure. Velocity vs. time data for the rocket example



Quadratic Interpolation (contd)

$$b_0 = v(t_0)$$

$$= 227.04$$

$$b_1 = \frac{v(t_1) - v(t_0)}{t_1 - t_0} = \frac{362.78 - 227.04}{15 - 10}$$

$$= 27.148$$

$$b_2 = \frac{\frac{v(t_2) - v(t_1)}{t_2 - t_1} - \frac{v(t_1) - v(t_0)}{t_1 - t_0}}{t_2 - t_0} = \frac{\frac{517.35 - 362.78}{20 - 15} - \frac{362.78 - 227.04}{15 - 10}}{20 - 10}$$

$$= \frac{30.914 - 27.148}{10}$$

$$= 0.37660$$

$$v(t) = b_0 + b_1(t - t_0) + b_2(t - t_0)(t - t_1)$$

$$= 227.04 + 27.148(t - 10) + 0.37660(t - 10)(t - 15), \quad 10 \leq t \leq 20$$

At $t = 16$,

$$v(16) = 227.04 + 27.148(16 - 10) + 0.37660(16 - 10)(16 - 15) = 392.19 \text{ m/s}$$

The absolute relative approximate error $|\epsilon_a|$ obtained between the results from the first order and second order polynomial is

$$|\epsilon_a| = \left| \frac{392.19 - 393.69}{392.19} \right| \times 100$$

$$= 0.38502 \%$$

General Form

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

where

$$b_0 = f[x_0] = f(x_0)$$

$$b_1 = f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_2 = f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

Rewriting

$$f_2(x) = f[x_0] + f[x_1, x_0](x - x_0) + f[x_2, x_1, x_0](x - x_0)(x - x_1)$$

General Form

Given $(n + 1)$ data points, $(x_0, y_0), (x_1, y_1), \dots, (x_{n-1}, y_{n-1}), (x_n, y_n)$ as

$$f_n(x) = b_0 + b_1(x - x_0) + \dots + b_n(x - x_0)(x - x_1)\dots(x - x_{n-1})$$

where

$$b_0 = f[x_0]$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

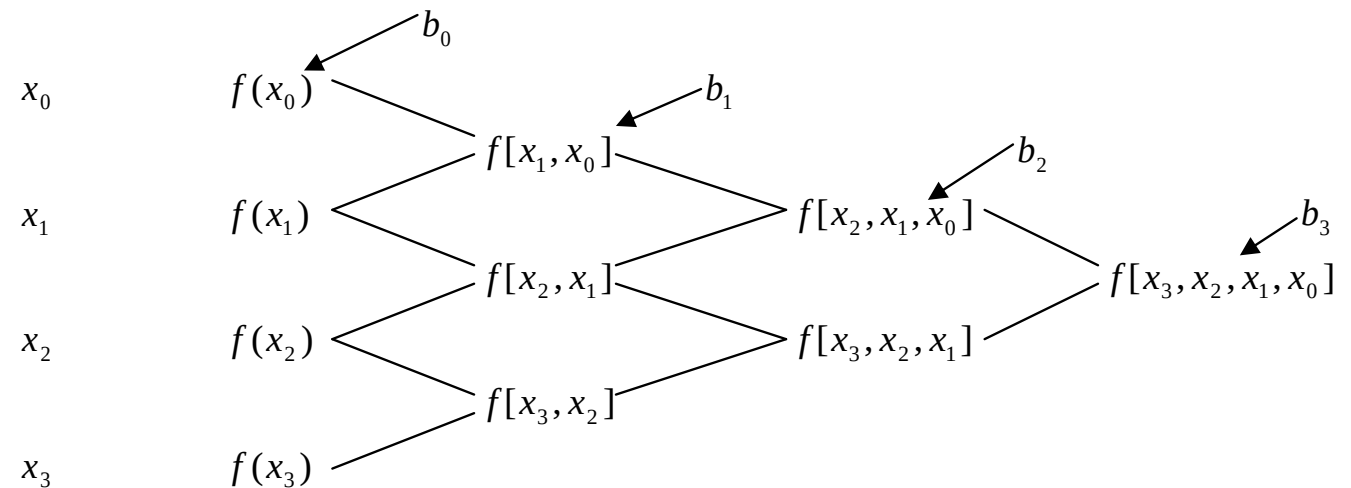
$\vdots$

$$b_{n-1} = f[x_{n-1}, x_{n-2}, \dots, x_0]$$

$$b_n = f[x_n, x_{n-1}, \dots, x_0]$$

The third order polynomial, given $(x_0, y_0), (x_1, y_1), (x_2, y_2)$, and (x_3, y_3) , is

$$f_3(x) = f[x_0] + f[x_1, x_0](x - x_0) + f[x_2, x_1, x_0](x - x_0)(x - x_1) + f[x_3, x_2, x_1, x_0](x - x_0)(x - x_1)(x - x_2)$$



Example

The upward velocity of a rocket is given as a function of time in Table 1. Find the velocity at $t=16$ seconds using the Newton Divided Difference method for cubic interpolation.

Table 3. Velocity as a function of time

t (s)	$v(t)$ (m/s)
0	0
10	227.04
15	362.78
20	517.35
22.5	602.97
30	901.67

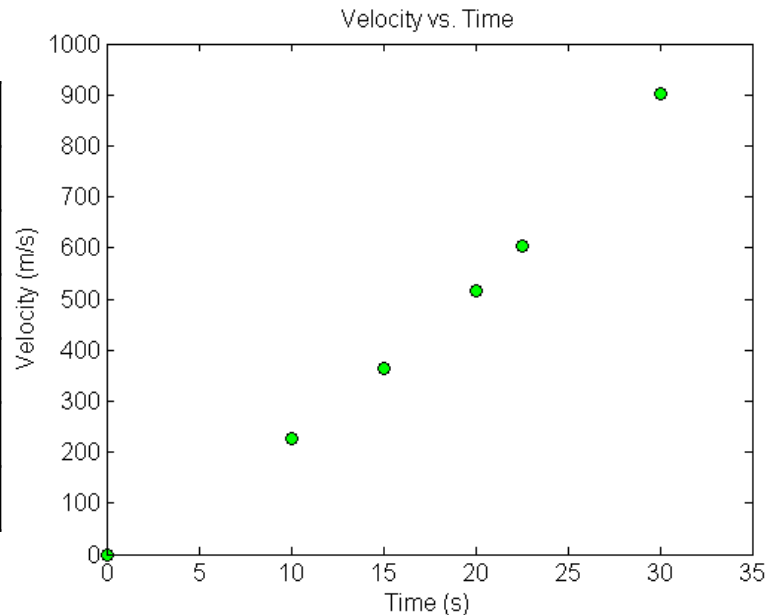


Figure. Velocity vs. time data for the rocket example

The velocity profile is chosen as

$$v(t) = b_0 + b_1(t - t_0) + b_2(t - t_0)(t - t_1) + b_3(t - t_0)(t - t_1)(t - t_2)$$

Example

we need to choose four data points that are closest to $t = 16$

$$t_0 = 10, \quad v(t_0) = 227.04$$

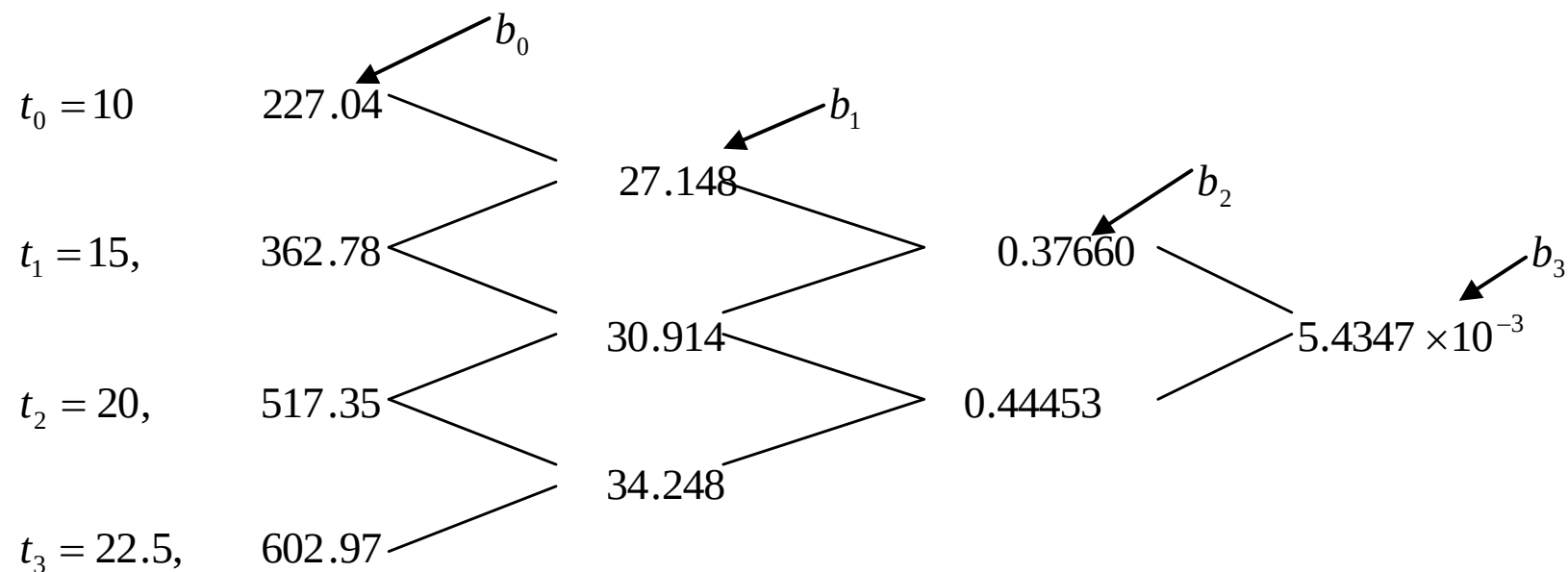
$$t_1 = 15, \quad v(t_1) = 362.78$$

$$t_2 = 20, \quad v(t_2) = 517.35$$

$$t_3 = 22.5, \quad v(t_3) = 602.97$$

The values of the constants are found as:

$$b_0 = 227.04; \quad b_1 = 27.148; \quad b_2 = 0.37660; \quad b_3 = 5.4347 \times 10^{-3}$$



Example

Hence

$$\begin{aligned} v(t) &= b_0 + b_1(t - t_0) + b_2(t - t_0)(t - t_1) + b_3(t - t_0)(t - t_1)(t - t_2) \\ &= 227.04 + 27.148(t - 10) + 0.37660(t - 10)(t - 15) \\ &\quad + 5.4347 * 10^{-3}(t - 10)(t - 15)(t - 20) \end{aligned}$$

At $t = 16$,

$$\begin{aligned} v(16) &= 227.04 + 27.148(16 - 10) + 0.37660(16 - 10)(16 - 15) \\ &\quad + 5.4347 * 10^{-3}(16 - 10)(16 - 15)(16 - 20) \\ &= 392.06 \text{ m/s} \end{aligned}$$

The absolute relative approximate error $|\epsilon_a|$ obtained is

$$\begin{aligned} |\epsilon_a| &= \left| \frac{392.06 - 392.19}{392.06} \right| \times 100 \\ &= 0.033427 \% \end{aligned}$$

Comparison Table

Order of Polynomial	1	2	3
$v(t=16)$ m/s	393.69	392.19	392.06
Absolute Relative Approximate Error	-----	0.38502 %	0.033427 %

Example

Problem Statement. Fit a second-order polynomial to the three points

$$x_0 = 1 \quad f(x_0) = 0$$

$$x_1 = 4 \quad f(x_1) = 1.386294$$

$$x_2 = 6 \quad f(x_2) = 1.791759$$

Use the polynomial to evaluate $\ln 2$.

The use of quadratic interpolation to estimate $\ln 2$. The linear interpolation from $x = 1$ to 4 is also included for comparison.

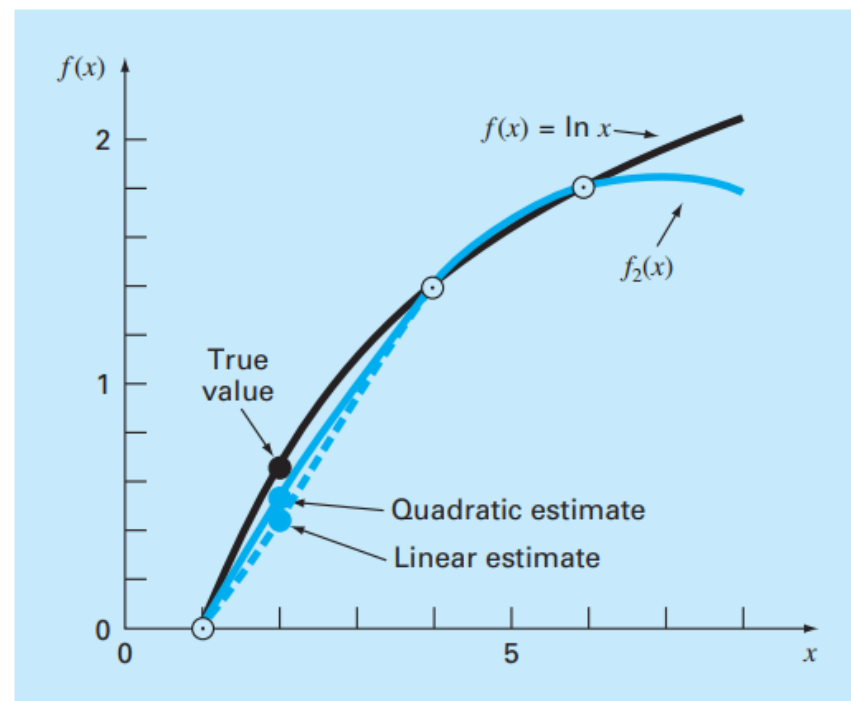
Solution

$$b_0 = 0$$

$$b_1 = \frac{1.386294 - 0}{4 - 1} = 0.4620981$$

$$b_2 = \frac{\frac{1.791759 - 1.386294}{6 - 4} - 0.4620981}{6 - 1} = -0.0518731$$

Substituting these values yields the quadratic formula $f_2(x) = 0 + 0.4620981(x - 1) - 0.0518731(x - 1)(x - 4)$ which can be evaluated at $x = 2$ for $f_2(2) = 0.565844$



Example

Now, adding a fourth point [$x_3 = 5$; $f(x_3) = 1.609438$], in the previous example, estimate $\ln 2$ with a third-order Newton's interpolating polynomial.

Solution

The third-order polynomial, with $n = 3$, is

$$f_3(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2)$$

The **first divided differences** for the problem are

$$f[x_1, x_0] = \frac{1.386294 - 0}{4 - 1} = 0.4620981$$

$$f[x_2, x_1] = \frac{1.791759 - 1.386294}{6 - 4} = 0.2027326$$

$$f[x_3, x_2] = \frac{1.609438 - 1.791759}{5 - 6} = 0.1823216$$

The **second divided differences** are

$$f[x_2, x_1, x_0] = \frac{0.2027326 - 0.4620981}{6 - 1} = -0.05187311$$

$$f[x_3, x_2, x_1] = \frac{0.1823216 - 0.2027326}{5 - 4} = -0.02041100$$

The **third divided difference** is

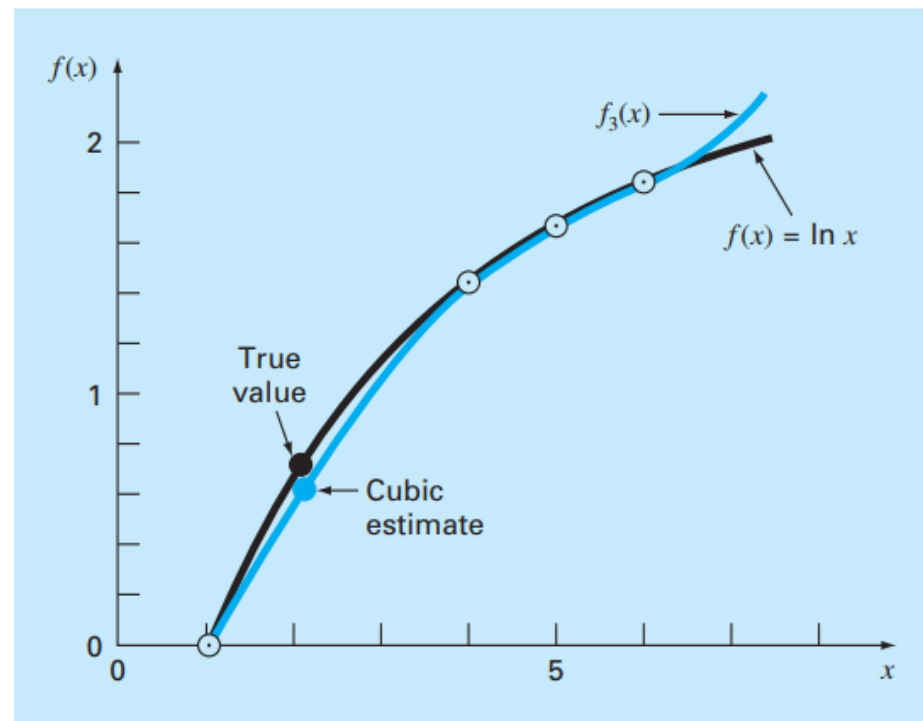
$$f[x_3, x_2, x_1, x_0] = \frac{-0.02041100 - (-0.05187311)}{5 - 1} = 0.007865529$$

Example

The results for $f[x_1, x_0]$, $f[x_2, x_1, x_0]$, and $f[x_3, x_2, x_1, x_0]$ represent the coefficients b_1 , b_2 , and b_3 , respectively, Along with $b_0 = f(x_0) = 0.0$.

$$f_3(x) = 0 + 0.4620981(x - 1) - 0.05187311(x - 1)(x - 4) + 0.007865529(x - 1)(x - 4)(x - 6)$$

$$f_3(2) = 0.6287686$$



Differences Between Lagrange Interpolation and Newton's Divided Difference Interpolation

Aspect	Lagrange Interpolation	Newton Divided Difference Interpolation
Definition	Constructs the interpolating polynomial directly using Lagrange basis polynomials .	Builds the interpolating polynomial incrementally using divided difference coefficients .
Computation Approach	Must be recomputed entirely if a new data point is added.	Can extend polynomial easily if a new data point is added (no need to start over).
Computational Efficiency	Simple for small datasets, but expensive for large datasets due to repetitive recalculation.	More efficient for large datasets; uses tables to compute divided differences only once.
Numerical Stability	Less stable for large number of points; may suffer from rounding errors.	More stable and preferred for practical numerical computation.
Preferred When	Fixed data points and polynomial needed in explicit form.	Data points may change, or incremental updates are required.
Ease of Implementation	Conceptually straightforward; a direct formula.	Slightly more complex but computationally superior.
Best Used For	Small datasets, hand calculation, theoretical understanding.	Larger datasets, iterative data collection, computer algorithms.

Practical Applications

Scenario / Real-Life Application	Why Lagrange Interpolation?	Why Newton Divided Difference?
Physics experiment with fixed readings (e.g., interpolate force vs. displacement for a lab report)	Data is fixed, and a single polynomial expression is needed for derivation or theoretical analysis.	— Not required since no new data points will be added.
Sensor data that updates over time (e.g., temperature readings every hour)	— Not efficient because full recalculation required for each additional data point.	Ideal because new readings can be appended easily using divided difference table.
Computer graphics curve fitting (interpolation for a fixed set of points)	Works well for generating one-time polynomial curves.	— Data rarely changes; no benefit of Newton updates.
Weather prediction models (data updated continuously)	— Lagrange would require complete recomputation.	Newton form is preferred because new data can be added without redoing previous calculations.
Engineering calibration tables (engine efficiency vs. RPM, battery discharge curves)	Good for symbolic representation of relationship in formula form.	Better when re-calibrations occur or extra measurements are included later.